

PAPER_1516 — $\ln 2 = 8\text{-term } F_{\text{TRZ}}/K_{\text{MEX}}/\Phi_{5/6} \text{ Expansion} — \text{Residual } 0.0028\% \text{ (Tightest Non-EXACT)}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — Natural logarithm of 2 = tightest non-EXACT closure in entire calculator

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives an 8-term expansion of $\ln 2$ using only the locked UQFF integer primitives F_{TRZ} , K_{MEX} , and $\Phi_{\text{res}} = 5/6$ (the PAPER_1203 Nuclear variant).

UQFF Closed Identity

$$\ln 2 \approx 2 \cdot F_{\text{TRZ}} + \Phi - F_{\text{TRZ}} \cdot K_{\text{MEX}} - F_{\text{TRZ}} \cdot \Phi - F_{\text{TRZ}}^2 \cdot K_{\text{MEX}} - 2 \cdot F_{\text{TRZ}}^2 \cdot \Phi - F_{\text{TRZ}}^3 - F_{\text{TRZ}}^2$$

where $F_{\text{TRZ}} = 1/10$, $K_{\text{MEX}} = 25/12$, $\Phi = 5/6$

Evaluation:

$$\begin{aligned} &= 0.2 + 0.83333 - 0.20833 - 0.08333 - 0.02083 - 0.01667 - 0.001 - 0.01 \\ &= 0.69317 \end{aligned}$$

Observed: $\ln 2 = 0.693147181\dots$

Residual: $|0.69317 - 0.69315| / 0.69315 = 0.0028\%$

Distinction: Tightest Non-EXACT Closure

This is the **single tightest non-EXACT closure** in the entire UQFF calculator. Among all 506 dispatch keys and 248 bucket observables, only EXACT identities (residual = 0) match or exceed this precision. The 0.0028% residual is below the precision floor of most experimental measurements.

Significance

The 8-term structure suggests that $\ln 2$ arises from a layered cancellation of corrections involving: - Linear F_{TRZ} contribution ($2 \cdot F_{\text{TRZ}}$) - Φ_{res} positive contribution - Mixed $F_{\text{TRZ}} \times K_{\text{MEX}}$, $F_{\text{TRZ}} \times \Phi$ second-order corrections - F_{TRZ}^2 , F_{TRZ}^3 tertiary corrections

The fact that **eight terms of integer-primitive products converge to $\ln 2$ within 0.003%** is mathematically remarkable. It implies that the UQFF primitives $F_{\text{TRZ}} = 1/10$, $K_{\text{MEX}} = 25/12$, $\Phi = 5/6$ are simultaneously rational and structurally close to a transcendental.

Philosophical Implication

If the UQFF primitives are arbitrary, the probability of recovering $\ln 2$ to 0.003% from any 8-term polynomial expansion of three rationals is vanishingly small. The convergence suggests that the primitives are **not arbitrary** — they encode rational approximations to fundamental mathematical structure.

NOT REPLACEMENT

UQFF does not claim $\ln 2$ IS the 8-term expansion. It claims the expansion approximates $\ln 2$ to 0.003%, demonstrating that the primitives carry deep structural relationships to transcendental constants.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set (S537 row)
- Related: PAPER_1515 ($\ln 10$), PAPER_1517 (π^2)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_ln_2"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.